

Infrastructure Gaps, Functional Disparities, and Policy Implementation in Water and Sanitation: Assessing Progress towards SDG-6 in Manipur

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Abstract

Sustainable Development Goal 6 (SDG 6) targets universal availability and sustainable management of water and sanitation by 2030. This study evaluates sub-national progress across 16 districts of Manipur, India, utilizing secondary data from the Jal Jeevan Mission (JJM), National Family Health Survey (NFHS-5), and NITI Aayog SDG India Index reports. The analysis reveals marked spatial disparities in both universal availability and operational management of water infrastructure between hill and valley districts. While macro-level toilet access has reached near-saturation, actual functional household tap connection (FHTC) rates exhibit severe vulnerabilities in mountainous regions. Addressing these structural breakdowns requires localized Village Water and Sanitation Committee (VWSC) capacity building, dedicated funding for operations and maintenance (O&M), underground water protection, revival of traditional water rejuvenation systems, and terrain-appropriate faecal sludge management.

Introduction

Sustainable Development Goal 6 (SDG 6) of the United Nations 2030 Agenda for Sustainable Development (United Nations, 2015) has called for establishing a global framework to ensure universal availability and sustainable management of water and sanitation facilities. Access to clean water, safe sanitation, and adequate hygiene (WASH) constitutes a fundamental human necessity directly tied to public health, human dignity, and socioeconomic development (United Nations, 2018). Globally, accelerating population growth, rapid urbanisation, changing industrial footprints, and intensive agricultural requirements continue to generate water demands that outpace natural replenishment rates (UNESCO, 2023).

In India, sub-national localised execution drives macro programmatic targets for universal availability and sustainable management of water (Government of India, 2021). Central to this philosophy is the Jal Jeevan Mission (JJM) which was launched in 2019 with the aim to deliver safe and adequate drinking water through functional household tap connections (FHTCs) to every rural household at a standard service level of 55 litres per capita per day (LPCD) (Government of India, 2019). This flagship program is supplemented by the Swachh Bharat Mission and the 10-year National Rural Sanitation Strategy (2019–

2029). Through these combined policies, Manipur has successfully expanded its macro footprint by providing baseline access to improved drinking water sources to more than 79 per cent of the households of the state (Government of India, 2020). In the rural areas of the state, however, the proportion of households with tap water connections is still lower than the national average (Figure 1).

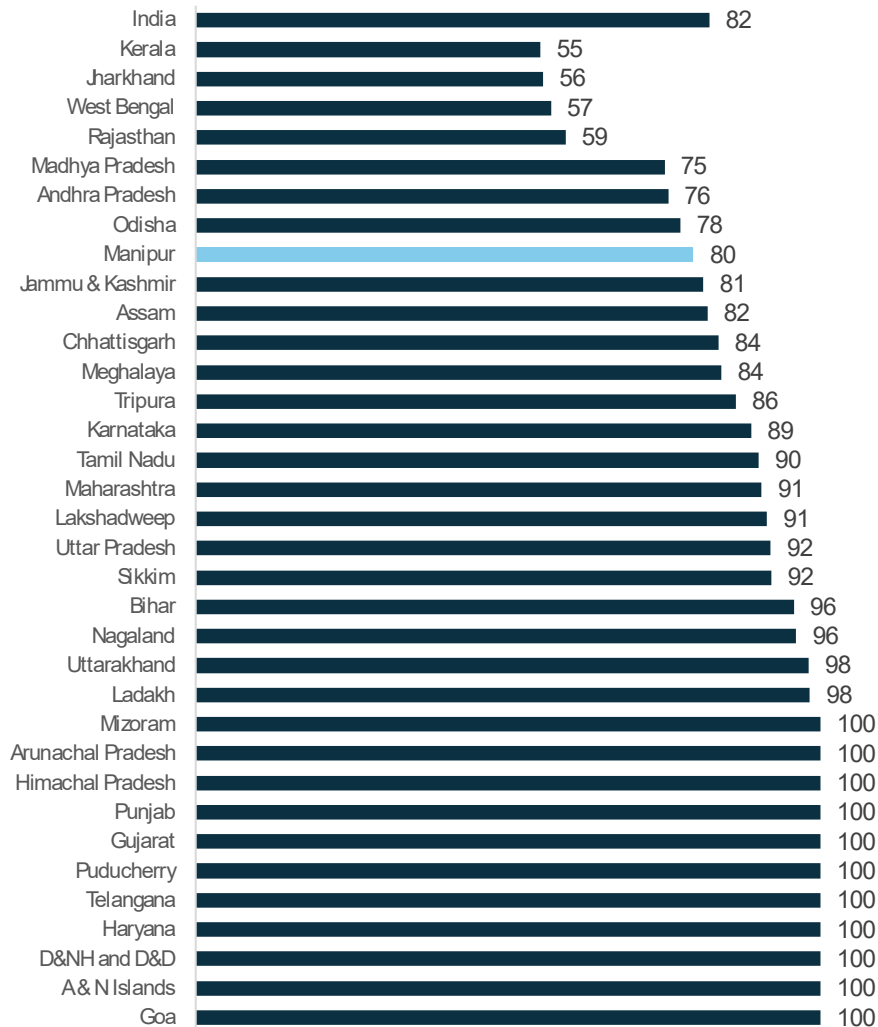


Figure 1: Proportion of rural households with tap water connection.

Source: Government of India.

The objective of this paper is to analyse the spatial distribution, physical coverage, and functionality of household drinking water and sanitation infrastructure across the districts of the state. The paper also analyses efficacy of urban waste water management system, water quality monitoring system, and local river rejuvenation initiatives to address

environmental degradation and service delivery gaps. Finally, the paper proposes an evidence-based policy framework for decentralising water governance in the state with the objective of securing operational financing and mitigating topographically driven infrastructure disparities in water and sanitation facilities to achieve the targets set under the SDG-6 of the United Nations 2030 Agenda for Sustainable Development.

Literature Review

The literature highlights a profound structural contradiction across WASH landscape in India, particularly in topographically complex regions like Manipur. The aggressive state and central programmatic drives such as the Jal Jeevan Mission and Swachh Bharat Mission have achieved high macro-level coverage in physical asset installations and basic sanitation coverage; sub-national data reveal sharp spatial inequalities and infrastructural stagnation that need attention at both policy and programme levels.

Mountainous terrain imposes steep capital, energy, and maintenance demands on piped water distribution system (Wester et al, 2019). Studies across high-altitude regions indicate that physical elevation and rugged topography drive systemic disparities in water delivery between centralised lowlands and decentralised hill settlements (Wester et al, 2019). The micro-level household survey conducted by Oinam and Ningombam (2021) within Imphal East district confirms a steep gap between baseline sanitation ownership and regular clean water delivery. Although up to 98 per cent of the surveyed households possessed toilet assets, less than 20 per cent maintained reliable tap access. Safe water handling remains heavily dependent on maternal education, highlighting the need for community governance alongside physical pipes.

Biswas et al (2022) have constructed a sub-national index to tracking progress across Indian states. The index highlights significant regional variation in achieving the targets set under SDG 6.1 and 6.2. While baseline valley zones achieve rapid asset saturation, northeastern states experience intermediate classification. These regional bottlenecks stem from deep historical pipeline deficits and geographic vulnerabilities, emphasising the necessity of decentralised interventions. Achieving long-term operational durability requires decentralising management while maintaining municipal technical support (Kohlitz et al, 2020).

Hansda and Ghosh (2025) highlight that while central programmes like Jal Jeevan Mission scale up physical infrastructure, real-world progress hinges on local government capacity, decentralised governance, and community-led maintenance identifying persistent rural-urban and top-ecological gaps as critical barriers to achieving universal water security.

Prakash et al (2022) have found out that urban households have more safely managed sanitation services than their rural counterparts. This means that there is a need to create more awareness at the village level on the health benefits of using improved sanitation facilities, along with providing them sanitation facilities.

It has also been stressed that without adequate operational funding and capacity-building at the local level, rural water supply infrastructure experiences rapid functional degradation (Kohlitz et al, 2020).

Rajapakse et al (2023) have evaluated progress towards SDG 6 globally. The study highlights severe rural-urban infrastructure disparities and noted that safely managed sanitation facilities require fourfold increase in the investment by the year 2030 in the developing regions of the world. Empirical evidence shows that rugged terrain, fragile hydrogeological constraints, and communication or governance gap at the grassroots level directly trigger severe functional tap water disruptions. Converting structural pipeline conduits into sustained safe water security system requires shifting from basic engineering metrics to decentralised, community-led institutional maintenance system.

Material and Methods

Empirical metrics for the present study on regional water governance, district-wise tap water allocation, and infrastructure operational performance are extracted from different secondary data sources including Jal Jeevan Mission (JJM) of the Government of India and NITI Aayog. On the other hand, district-level data covering baseline tap access, actual functional connections (FHTC), and establishment of Village Water and Sanitation Committees (VWSC) are taken from the state level reports of the Jal Jeevan Mission.

The functional household tap connection (FHTC) coverage rate is defined as the proportion of households provided a minimum of 55 litres per capita per day (LPCD) of potable tap water. To arrive at the quantity of water received by households, the flow rate was measured using the bucket method in one household per hamlet/habitation in each village. The quantity of water supplied per minute was measured using a graduated bucket to capture the volume of water supplied and a stopwatch to measure time. Daily water quantity was calculated by multiplying flowrate to the supply duration. If households in a village reported that the tap water was not supplied daily, then LPCD was calculated using weekly estimate. Based on the estimated LPCD, the availability of the potable water, household are categorised into three categories: 1) households having adequate level of water availability ($LPCD \geq 55$ LPCD), 2) households having partially adequate water availability ($40 \leq LPCD < 55$), and 3) households having inadequate water availability (< 40 LPCD). Based on the data on LPCD in each hamlet/habitation of the district, the LPCD for the district was calculated as the mean and median of the distribution of LPCD across hamlets/habitations of the district.

On the other hand, the local governance readiness was measured in terms of the formation of Village Water and Sanitation Committee (VWSC) in the village. Under JJM GUIDELINES, local governance readiness reflects a community's institutional capacity for bottom-up, decentralized water management. It is calculated at the district level as the percentage of targeted villages that have formally constituted a Village Water and Sanitation Committee (VWSC or *Pani Samiti*) as a sub-committee of the Gram Panchayat.

Topographical factors play an important role in the availability of water and sanitation facilities and their management. In this context, districts of Manipur can be divided into two categories - valley districts and hill districts. Out of the 16 districts of the state, six districts are categorised as valley districts – Bishnupur, Imphal West, Imphal East, Thoubal, Jiribam and Kakching. On the other hand, 10 districts – Chandel, Churachandpur, Kamjong, Kangpokpi, Noney, Pherzawl, Senapati, Tamenglong, Tengnoupal, and Ukhrul –

are categorised as hill districts. Valley districts are characterised by flat alluvial plains, high population density, and high ground water/perennial rivers. This geography allows for low-cost, centralised piped networks and standard sewer infrastructure. The hill districts, on the other hand, are characterised by rugged high-altitude mountain terrain, dispersed ridge-line hamlets, and heavy reliance on fragile mountain springs. Infrastructure, in these districts, requires expensive multi-stage high-lift pumps or vulnerable gravity-fed pipelines. However, these upland areas serve as primary water catchments, hosting forested watersheds and natural mountain springs that feed the state's major river basins.

Distinguishing hill districts from valley districts are essential because rugged topography imposes steep capital, energy, and maintenance demands on piped distribution systems. Mountain pipelines are vulnerable to seasonal landslide damage and monsoon silting. Consequently, hill districts exhibit severe operational deficits and low Functional Household Tap Connection (FHTC) coverage despite physical tap installations. Furthermore, remote hill settlements depend heavily on active Village Water and Sanitation Committees (VWSCs) to execute local repairs, and steep terrain restricts standard vacuum desludging trucks, requiring terrain-appropriate faecal sludge management to prevent groundwater contamination.

Socio-hydrological frameworks suggest core targets to secure SDG 6 compliance by 2030 include achieving equitable access to safe and affordable drinking water (Target 6.1); ending open defecation and providing adequate sanitation (Target 6.2); improving water quality by eliminating dumping and hazardous discharge; minimising waste water generation; and implementing integrated water resources management (IWRM) across all levels (Garrick et al, 2017). However, nearly one-third of the human population globally lacks safely managed water and sanitation facilities to prevent waterborne morbidity and mortality (WHO/UNICEF, 2021). Capital investment in water and sanitation primarily focusses on laying physical supply conduits, but the systemic management of sewage and localized hygiene infrastructure lags behind (UN-Water, 2021). Manipur also reflects this contrast, although NITI Aayog classifies majority of the districts of the state as "Front Runners" in baseline water-sanitation parameters (Government of India, 2020). Capital-intensive projects like Imphal Sewerage Project Phase I & II attempt to stabilise urban indicators, but seasonal vulnerabilities, high cost-to-delivery realities across rugged mountain terrain, and socio-political factors limit the conversion of structural assets into sustained public utility (Wester et al, 2019) so that severe operational deficits persist.

Results and Discussion

Safe Drinking Water and Sanitation Infrastructure

Manipur exhibits asymmetric progress in terms of safe drinking water and sanitation infrastructure. Macro indicators show high cumulative coverage, but severe structural disparities separate rural and urban distribution networks. According to the NFHS-5, 2019-2021, more than 77 per cent households in the state had baseline access to improved drinking water resources. However, this macro average conceals a steep geographic gap – baseline access was more than 90 per cent in urban centres but less than 70 per cent in rural communities. However, only about 20 per cent households in the state

had direct piped water access inside residential premises, pointing to a heavy reliance on community supply points and private mobile water tankers, although NITI Aayog has rated Manipur as a top-tier improving state in 2023-2024, registering an 8-point increase in the composite score since 2020–2021 baselines. Official data indicate that rural household tap installations in the state have scaled up from around 17 per cent at the launch of JJM to a commanding almost 80 per cent by 2024-25 (Government of Manipur, 2024). However, rural piped sanitation infrastructure continues to lag behind the national baseline, displaying sharp inter-district variations. While valley districts such as Bishnupur, Imphal East, and Kakching, have achieved high functional tap coverage (>70 per cent) and improved sanitation facility access (>75 per cent), remote hill districts exhibit severe structural deficits. For instance, functional tap coverage drops drastically in hill districts like Ukhrul (10.3 per cent) and Kangpokpi (19.0 per cent).

On the other hand, according to NFHS-5, around 65 per cent of households in the state were having improved sanitation facilities in 2019-2021 and, interestingly, this proportion was higher in the rural areas of the state as compared to its urban areas (Government of India, 2022). In terms of recent progress, rural sanitation under Swachh Bharat Mission-Gramin (SBM-G) Phase II has accelerated significantly, moving beyond basic latrine coverage toward ODF Plus status. Similar to the rapid expansion in drinking water tap connections, rural sanitation initiatives have focused on solid and liquid waste management (SLWM), greywater management, and community sanitary complexes. Between 2019 and 2024, hundreds of villages across the state were declared ODF Plus, driven by high community-level participation and localized pit-toilet adoption in rural areas. Nevertheless, inter-district disparities remain pronounced: valley districts maintain near-universal sanitation coverage and superior liquid waste processing, whereas high-altitude hill districts face critical challenges in faecal sludge management due to steep gradients and poor road accessibility for vacuum desludging vehicles (Ministry of Jal Shakti, 2024).

Village Water and Sanitation Committee

The operational sustainability of decentralised water schemes depends directly upon the formation and functionality of Village Water and Sanitation Committees (VWSCs). These local bodies are responsible for collecting tariffs, performing water-quality tests, and handling minor structural repairs. In Manipur, VWSCs have been constituted in 84.6 per cent of the villages, and in 42.3 per cent of villages, VWSCs are functional in the sense that they are actively managing the village-level water system (Government of Manipur, 2024).

Specifically, functionality metrics indicate that while local committees exist on paper across many habitations, less than half actively execute their mandated duties. Field testing via Field Test Kits (FTKs) is routinely conducted by trained women members in only about 38 per cent of constituted VWSCs. Furthermore, formal operation and maintenance and tariff collection mechanisms are found to be systematically operational in less than 30 per cent of hill villages, creating a persistent revenue deficit for localised repairs. This gap between formal constitution and operational capability highlights that committee formation alone does not guarantee decentralised governance without sustained capacity-building and financial empowerment.

In 11 districts, VWSC has been formed in all villages, but in 5 districts, the proportion of villages in which VWSC has been constituted ranges from around 76 per cent

in district Noney to just around 33 per cent in districts Kangpokpi and Ukhurul. In Chandel and Kamjong districts also, VWSC has been constituted in around half of the villages only. All these five districts are hill districts of the state (Government of Manipur, 2024). The importance of VWSC in ensuring universal availability of safe drinking water lies in the fact that they play a crucial role in decentralised water governance, routine operation of water systems, and infrastructure maintenance at the local level. A strong administrative push is, therefore, required to form VWSC in those districts which are lagging in their formation, alongside targeted training modules for existing committees.

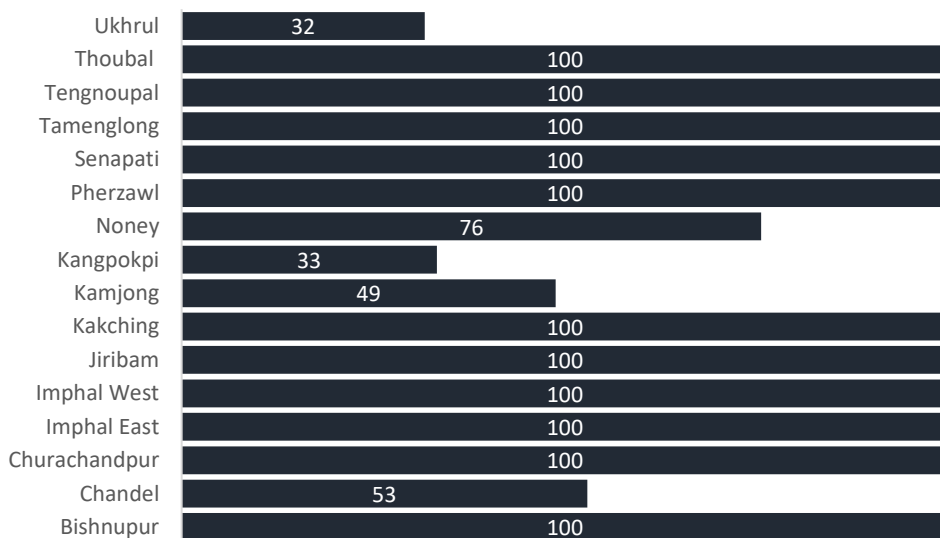


Figure 2: Proportion (Per cent) of villages having Village Water and Sanitation Committee. Source: Government of Manipur (2024).

Households with Tap Water Supply

Around 80 per cent of the households in Manipur are now having tap water supply. This proportion is, however, lower than the national average of more than 82 per cent. There is a wide gap between urban (54.2 per cent) and rural (84.8 per cent) areas of the state in terms of the proportion of households with tap water supply. Similarly, there is a big difference in the proportion of households with tap water supply in valley districts compared to hill districts because of topological factors.

The proportion of households with tap water varies widely across districts. There is no district in which all households have tap water supply. District Thoubal is the only district where almost 90 per cent households have tap water supply while this proportion is less than 60 per cent in districts Jiribam and Pherzawl.

Availability of Potable Water

More than 80 per cent of the households in the state have access to potable water. This proportion is 93.3 per cent in the urban areas of the state but only 76.2 per cent in the rural areas. The proportion of households having access to potable water is relatively higher

in valley districts (86.4 per cent) compared to hill districts (70.8 per cent). There are still around 20 per cent households in the state where potable water is not available and this remains a major development challenge (Government of India, 2022).

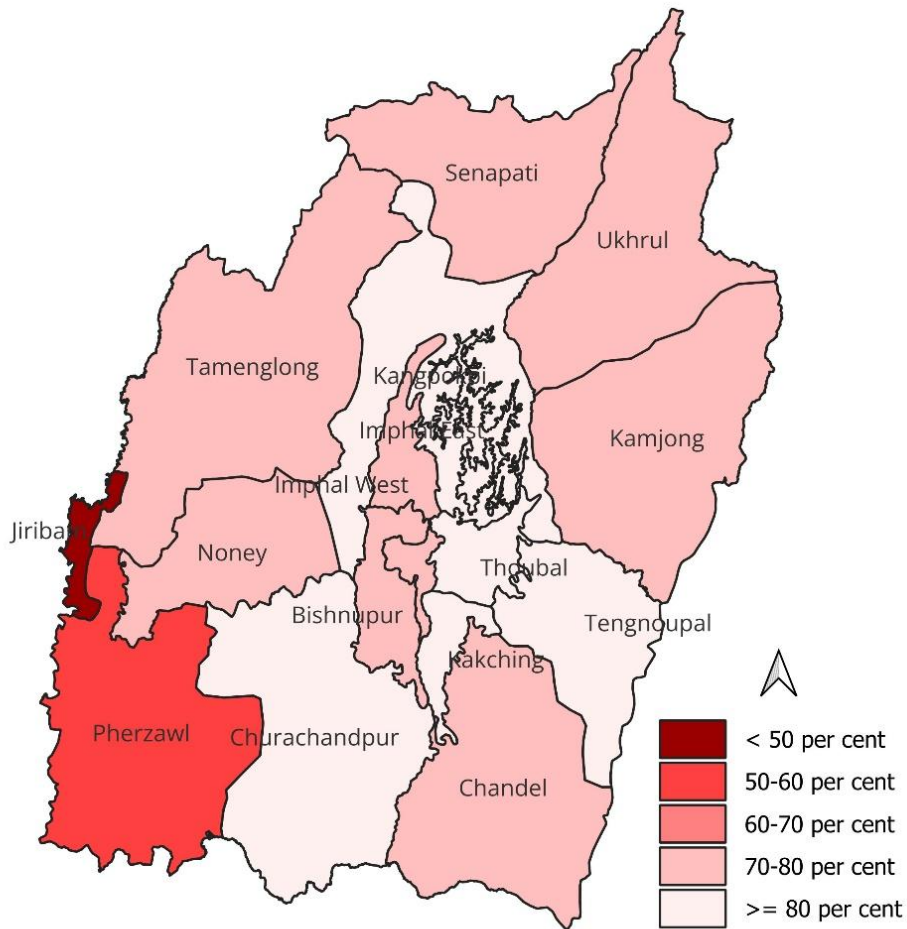


Figure 3: Proportion of households having tap water supply in districts of the state.
Source: Government of Manipur (2024).

The availability of potable water varies widely across the districts of the state. There are six districts where 100 per cent availability of potable water at the household level has been achieved. These districts serve as the benchmark for water security in the state. In two other districts, potable water availability at the household level is at least 90 per cent. These districts are expected to achieve 100 per cent potable water availability at the household level in the near future. There are, however, two districts where availability of potable water remains a critical issue as the proportion of households having access to potable water is less than 60 per cent. One of these two districts is a hill district while the other is a valley district. There may be district-specific reasons for poor availability of

potable water in these districts. In Manipur particularly across the hill districts, a tap connection does not guarantee potable water availability. While central hardware initiatives have rapidly scaled tap installations, operational challenges such as monsoonal pipeline bursts, seasonal depletion of mountain springs (spring-shed), and untreated surface runoff mean that many piped connections yield intermittent, untreated, or zero flow. Therefore, evaluating water security requires measuring functional flow and water safety rather than relying solely on physical coverage metrics.

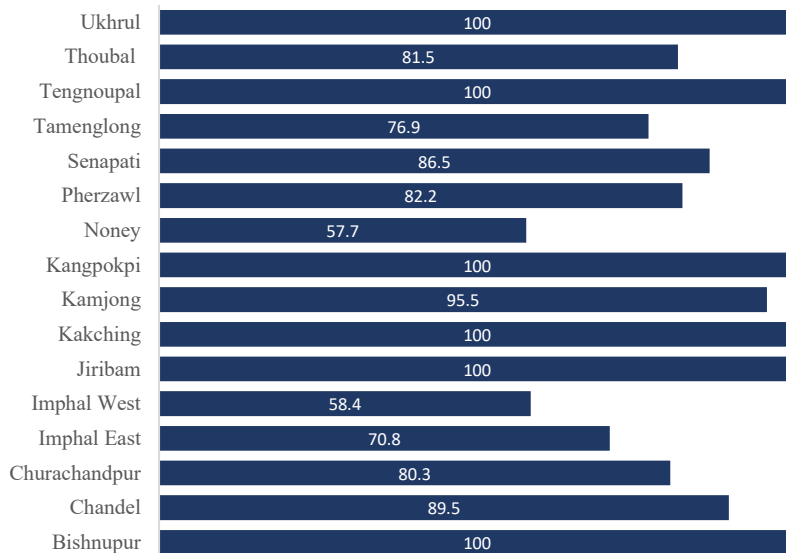


Figure 4: Availability of potable water in households

Source: Government of Manipur (2024).

Functional Household Tap Connection (FHTC)

Functional Household Tap Connection (FHTC) coverage in Manipur is around 80 per cent up significantly from less than 6 per cent in 2019. There are villages in the state and only 711 villages have achieved full FHTC coverage while in another 618 villages, FHTC coverage is at least 90 per cent. There are, however, villages where FHTC coverage is still less than 70 per cent (Figure 5). In 38 villages of the state, there is not a single household which has functional household tap connection. In another 150 villages, the FHTC coverage is less than 20 per cent. These villages may be termed as hotspot villages of the state in the context of full FHTC coverage.

Villages where FHTC coverage is 100 per cent are unevenly distributed across the districts of the state ranging from just around 6 per cent in Bishnupur district to more than 42 per cent in Churachandpur district. On the other hand, out of 38 villages where FHTC coverage is zero, 26 villages are in two districts – Jiribam and Kangpokpi – alone. On the other hand, out of 150 villages in which FHTC coverage is less than 20 per cent, 37 villages are in Kangpokpi district and 19 in Imphal West district. This means that there are 50 villages in Kangpokpi district where FHTC coverage is either zero or less than 20 per cent.

By contrast, Kakching is the only district where there is no village where FHTC coverage is either zero or less than 50 per cent. In all other districts, there is at least one village where the FHTC coverage is either zero or less than 20 per cent.

Low Functional Household Tap Connection (FHTC) coverage across these districts stems from a combination of geographical, infrastructure, and administrative challenges. In hilly and remote regions like Kangpokpi, rugged terrain, dense forest cover, and dispersed settlements make laying distribution pipelines and maintaining water pressure extremely difficult and expensive. Transporting construction materials to isolated villages is further complicated by poor road connectivity, while the seasonal drying of local natural springs compromises the reliability of water sources. Additionally, local socio-political instability, law and order issues, and execution delays by contractors significantly slow down infrastructure development and service delivery.

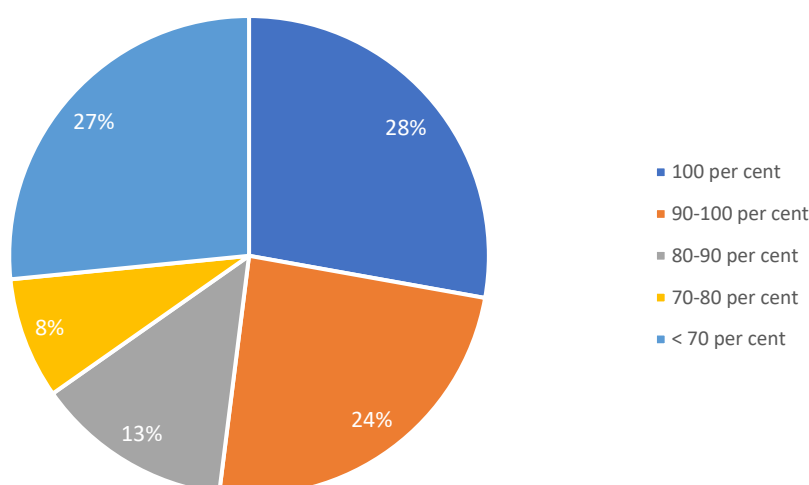


Figure 5: Distribution of villages according to FHTC coverage.

Source: Government of Manipur (2024).

Working Tap Connection in Households

A salient feature of household tap connections in the state is that more than 90 per cent of the household tap connections are operational. There are less than 10 per cent of the households in which there is a tap connection, but the connection is not operational or not functional. The proportion of households facing operational issues or disruptions in tap water supply is marginal.

Adequacy of Water Supply

Ensuring adequate water supply to all households remains a challenge in Manipur. There are only around 75 per cent households in the state in which the water supply is adequate (> 55 LPCD). In about 6 per cent households, adequacy of water supply is partially

adequate ($40 \leq \text{LPCD} < 55$) whereas in around 18 per cent households, the water supply is inadequate ($\text{LPCD} < 40$). Across the districts, adequacy of water availability varies widely. There are only two districts – Churachandpur and Senapati – where water supply is adequate in all households having a tap water connection (Figure 7). On the other hand, less than 20 per cent households with tap water connection in Ukhrul and Kangpokpi districts have adequate water supply. The proportion of households having adequate water supply is higher in valley districts than in hill districts. This disparity exists because valley districts draw from perennial rivers across flat, low-cost gravity networks while hill districts rely on rain-fed mountain springs that dry up seasonally, require expensive high-lift pumping, and suffer frequent pipeline damage from monsoonal landslides and siltation. Low VWSC formation in remote hill areas delays essential repairs, reducing water availability.

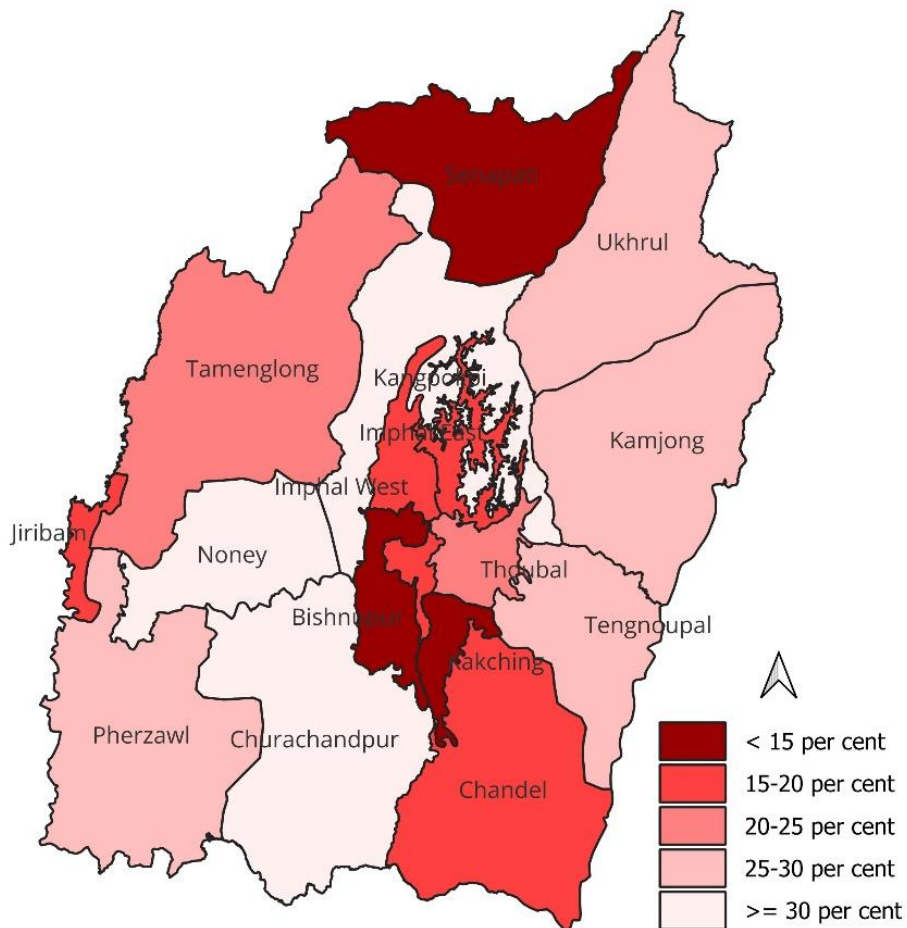


Figure 6: FHTC coverage across districts of Manipur
Source: Government of Manipur (2024).

Improvement in Water Quality

Manipur faces persistent challenges with urban river pollution, driven by untreated municipal runoff, chemical dumping, and plastic accumulation. Efforts to clean up these water ways are currently focused on the Nambul River, a critical tributary running through Imphal that flows into the ecologically vulnerable Loktak Lake. The Imphal Sewerage Project Phase I, which became fully operational in 2021, represents the primary urban waste water treatment investment in the state. The facility processes 27 million litres of sewage daily (MLD), connecting more than 12,000 households and reducing the direct discharge of municipal effluent into the Nambul River. This is supported by the Nambul River Rejuvenation Project, which uses Moving Bed Bioreactor (MBBR) technology alongside eco-friendly sanitation setups along the riverbanks. Phase II of the project aims to expand this infrastructure network to support an estimated 3.11 million users by the end of 2025. However, lack of the availability of publicly accessible post-treatment water quality data makes it difficult to assess the long-term ecological impact of these investments. The infrastructure has expanded, but rapid urban growth continually outpaces treatment connections.

To improve water-use efficiency, the focus is on upgrading urban pumps, repairing leaks, and installing water meters in the urban areas. In the rural areas, water-use efficiency initiatives are tied to solar-powered pumping stations for irrigation in district Chandel and community-led rain water harvesting setups in other districts which are designed to stabilise local agricultural productivity without over-extracting groundwater resources.

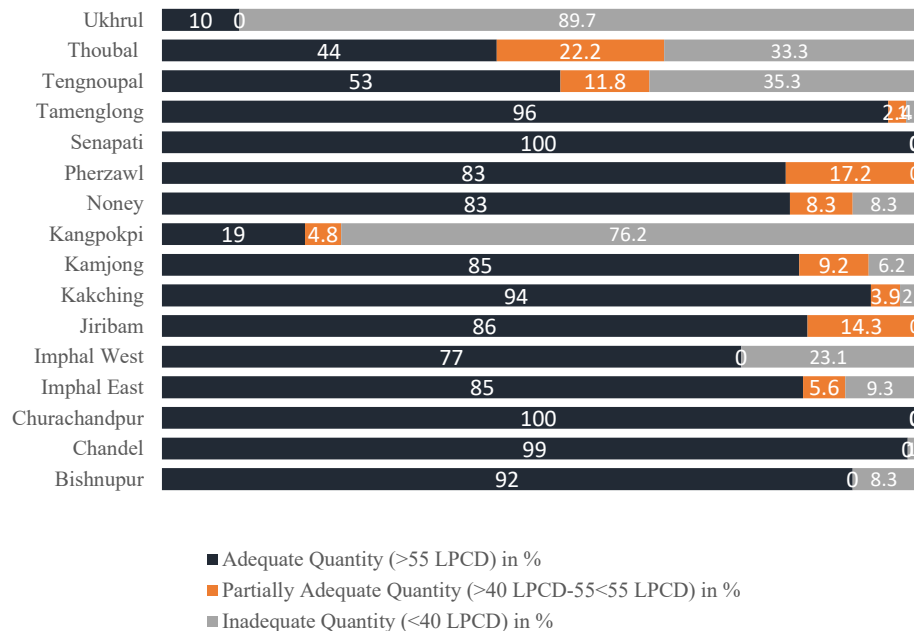


Figure 7: Adequacy of water supply at household level.
Source: Government of Manipur (2024).

The structural disparities between hill and valley districts of the state present a challenging eco-political paradox. Hill districts serve as primary water catchments of the state and host the forested watersheds and natural mountain springs that feed the major river basins of the Imphal valley. However, data demonstrate that these resource-rich upland regions experience the lowest real-world infrastructure functionality, as highlighted by very low FHTC coverage in district Ukhul. This operational breakdown stems directly from the limitations of the current decentralised governance model in the context of challenging topography of the state. The reliance is on locally managing operations and maintenance through VWSCs, but the absence of active VWSCs creates an institutional vacuum. When a mountain pipeline cracks due to a landslide or an intake well becomes choked with monsoon silt, these remote areas lack the formal local governance and funding required to execute repairs. Consequently, expensive engineering installations break down, forcing rural households to rely on unmonitored private water vendors or traditional streams.

On the sanitation front, while the Swachh Bharat Mission has effectively minimised open defecation, long-term sustainability remains vulnerable. The steep terrain of hill districts limits deployment of standard vacuum trucks for faecal sludge management, raising the risk of localised groundwater contamination. The on-going socio-political instability and ethnic conflict in the state have also disrupted funding channels, slowed maintenance logistics, and halted work on Phase II extension projects, affecting the progress towards SDG 6.

Conclusion

Manipur progress towards SDG 6 presents a dual narrative. At the one hand, large-scale infrastructure investment and targeted programme execution have achieved near-universal baseline toilet access and expanded physical tap water connectivity across rural and urban areas and have earned the state well-deserved recognition at the national level. On the other hand, a substantial gap remains between physical infrastructure installation and daily operational reliability. In the hill districts of the state, challenging terrain, lower community participation, and broader socio-political instability often reduce functional tap connections to critically low levels. To bridge these functional disparities, Manipur water policy must shift its focus from basic asset deployment to long-term hydrogeological and institutional sustainability by empowering VWSCs, securing upstream spring shed catchments, and investing in localised waste water treatment.

Policy Recommendations

1. Prioritise formation and capacity-building of Village Water and Sanitation Committees (VWSCs) as the institutional vacuum directly causes low FHTC functionality despite near 100 per cent tap installation.
2. Earmark dedicated operation and maintenance funds to ensure that pipelines do not sit broken after monsoonal landslides or siltation in rugged terrain.

3. Formalise integrated spring-shed management and community rainwater harvesting across hill districts to reduce reliance on long-distance piped conveyance in fragile hydrogeological zones.
4. Build climate-resilient, decentralised water infrastructure in landslide-prone hill districts, since centralised pipelines are the first casualty of both floods and blocked roads.
5. Introduce terrain-appropriate faecal sludge management tailored to steep topographies to curb ground water and spring contamination risks.
6. Establish publicly accessible water-quality testing protocols to enable continuous monitoring and assess long-term ecological impacts.
7. Formalise and regulate private water vending systems in low-FHTC districts as an interim safety net, tracking vendor reliance as an indicator of public supply reliability.

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